

BOX OFFICE

DATA SCIENCE REPORT



Avatar



Jurassic Park



Toy Story



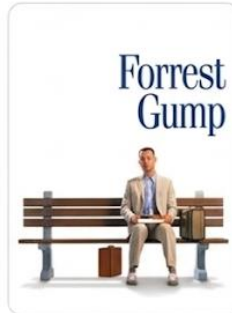
Minions



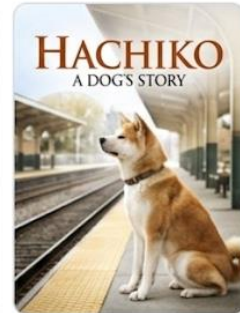
Mars Needs Moms



Gone with the Wind



Forrest Gump



Hachiko: A Dog's Story

by Kadir Arslan

Created: 25.12.2024

Last Edited: 19.03.2026

Movies Financial Performance Report

The film industry is a high-risk, high-reward business where production budgets can range from modest independent projects to massive blockbuster investments. To assess a film's financial success, it is essential to understand how budgets impact revenue, profitability, and long-term trends. In this report, we are going to examine a comprehensive dataset on films, focusing on production budgets, domestic and global box office incomes, and overall financial results.

Using Python-based data analysis tools, we are going to investigate how financial performance in the movie industry has evolved over time and what factors contribute to successful films. The analysis examines the relationship between budget and box office revenue, identifies profitable and unprofitable films, compares domestic box office revenue with global revenue, and assesses how profitability and investment strategies have evolved over the course of film history.

Using data cleaning, feature engineering, interactive visualizations, and regression analyses, this report aims to gain meaningful insights into the film industry and answer a key question: Do higher production budgets actually lead to greater box office success, or are other factors more decisive for a film's financial success?

Source Data Files (CSV Format)

Movies box office figures: github.com

Import lines:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from pandas.plotting import register_matplotlib_converters
from matplotlib.ticker import FuncFormatter
register_matplotlib_converters()
import matplotlib.ticker as ticker
import textwrap
```

Display Formatting for Financial Data

To improve readability and ensure consistent formatting of monetary amounts, we configure pandas to display floating-point numbers with two decimal places and a thousand separators.

```
# Notebook Presentation
pd.options.display.float_format = '{:,.2f}'.format
```

Acquiring the Source and Exploring Initial Dataset

Before we begin the analysis, we'll first load the dataset and examine its overall structure. This step gives us a quick overview of the dataset's size and allows us to check the first and last entries to better understand the available data.

```
data = pd.read_csv('csv_files/movies_box_office_figures.csv')

# How many rows and columns does the dataset contain?
print("\nShape of the data:")
print(data.shape)

print("\nFirst entries of the data:")
print(data.head())

print("\nLast entries of the data:")
print(data.tail())
```

Shape of the data:

```
(5391, 6)
```

First rows of the data:

	Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
0	5293	8/2/1915	The Birth of a Nation	\$110,000	\$11,000,000	\$10,000,000
1	5140	5/9/1916	Intolerance	\$385,907	\$0	\$0
2	5230	12/24/1916	20,000 Leagues Under the Sea	\$200,000	\$8,000,000	\$8,000,000
3	5299	9/17/1920	Over the Hill to the Poorhouse	\$100,000	\$3,000,000	\$3,000,000
4	5222	1/1/1925	The Big Parade	\$245,000	\$22,000,000	\$11,000,000

Last entries of the data:

	Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
5386	2950	10/8/2018	Meg	\$15,000,000	\$0	\$0
5387	126	12/18/2018	Aquaman	\$160,000,000	\$0	\$0
5388	96	12/31/2020	Singularity	\$175,000,000	\$0	\$0
5389	1119	12/31/2020	Hannibal the Conqueror	\$50,000,000	\$0	\$0
5390	2517	12/31/2020	Story of Bonnie and Clyde, The	\$20,000,000	\$0	\$0

The dataset contains 5,391 movies, where each row represents a single film and its financial performance.

From the dataset shape, we observe that there are six columns. The *Rank* column provides a unique identifier for each movie. *USD_Domestic_Gross* shows box office earnings in the United States and Canada, and *USD_Worldwide_Gross* reflects the total global revenue, including both domestic and international markets.

The dataset spans a long historical period and includes films released between 1915 and 2020, enabling an analysis of trends across more than a century of film history.

One notable observation is that some of the most recent entries show zero values for gross revenue. This is because the dataset was collected on May 1, 2018, meaning that some films with later release dates had not yet been released or had not yet generated box office revenue at the time of data collection.

Checking for Missing and Duplicate Data

Before beginning data cleaning and analysis, it is important to verify the integrity of the dataset by checking for missing values and duplicate entries.

If not properly addressed, these issues can skew statistical results and lead to incorrect conclusions.

```
# Are there any NaN values present?
print("\nIs There any NaN values in the data:")
print(data.isna().values.any())

# Are there any duplicate rows?
print("\nIs There any duplicated values in the data:")
print(data.duplicated().values.any())
```

```
Is There any NaN values in the data:
```

```
False
```

```
Is There any duplicated values in the data:
```

```
False
```

The inspection shows that the dataset contains no NaN (missing) values and no duplicated rows, meaning each movie entry is unique and complete.

This confirms that the dataset is structurally consistent and ready for further preprocessing steps.

Obtaining the Overview of the DataFrame and DataTypes

To better understand the structure of the dataset and the data types, we use the `info()` function, which provides a clear summary of the columns, the number of non-zero values, and the corresponding data types.

```
# Inspection of the dataframe  
print(data.info())
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 5391 entries, 0 to 5390  
Data columns (total 6 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   Rank                  5391 non-null  int64  
1   Release_Date          5391 non-null  str  
2   Movie_Title           5391 non-null  str  
3   USD_Production_Budget  5391 non-null  str  
4   USD_Worldwide_Gross    5391 non-null  str  
5   USD_Domestic_Gross     5391 non-null  str  
dtypes: int64(1), str(5)
```

The output shows that all 5,391 entries are non-zero, which confirms that the dataset is complete.

However, most columns are currently stored as strings (objects), including “Release_Date” and the three financial columns.

While this format is suitable for headings, the columns for release date, budget, and revenue should be adjusted to allow for calculations such as profit, margins, and statistical analyses.

Converting Columns to Appropriate Data Types

Before performing calculations and visualizations, the columns in the dataset must be converted to appropriate data types. In particular, the release dates of movies should be stored as a datetime object, and the financial columns should be converted from strings to numeric values to enable mathematical operations.

Converting Strings to Datetime Datatypes

```
# Convert the Release_Date column to a Pandas Datetime type.  
data.Release_Date = pd.to_datetime(data.Release_Date)
```

The “Release_Date” column is converted to the Pandas datetime format. This allows us to perform time-based analyses, such as grouping by year, examining trends over time, and comparing historical trends across different time periods.

Converting Strings to Numeric Datatypes

The financial columns contain non-numeric characters, such as dollar signs and commas, which prevent numerical evaluation. To resolve this issue, we remove these characters using a nested loop and then convert the cleaned values to a numeric format. This process ensures that these financial columns can be used for calculations such as profit, margins, correlations, and regression analyses.

```
chars_to_remove = [',', '$']  
  
columns_to_clean = ['USD_Production_Budget', 'USD_Worldwide_Gross',  
                    'USD_Domestic_Gross']
```

```

for col in columns_to_clean:
    for char in chars_to_remove:
        # Removing the unrelated characters:
        data[col] = data[col].astype(str).str.replace(char, "")
    # Now, we have just plain numbers.
    # We are converting these numbers to "numeric" datatype
    data[col] = pd.to_numeric(data[col])

```

Verification of Data Type Conversion

After performing the conversions, we are checking the DataFrame again using the `info()` function to confirm the changes

```
print(data.info())
```

```

<class 'pandas.DataFrame'>
RangeIndex: 5391 entries, 0 to 5390
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Rank                                  5391 non-null   int64
1   Release_Date                         5391 non-null   datetime64[us]
2   Movie_Title                          5391 non-null   str
3   USD_Production_Budget                5391 non-null   int64
4   USD_Worldwide_Gross                 5391 non-null   int64
5   USD_Domestic_Gross                  5391 non-null   int64
dtypes: datetime64[us](1), int64(4), str(1)

```

The output shows that the “Release_Date” column is now in Datetime format, while all financial columns have been successfully converted to integer types.

This confirms that the dataset is now correctly structured and ready for precise financial analysis, visualizations, and further feature engineering steps.

Descriptive Statistics of Budget and Revenue

To better understand the financial characteristics of the dataset, we calculate descriptive statistics for the columns “Production Budget,” “Worldwide Gross Revenue,” and “Domestic Gross Revenue.”

This summary includes key metrics such as the mean, minimum, maximum, and quartiles, allowing us to quickly assess general trends and fluctuations in film finances.

```
columns_to_analyze =  
    ["USD_Production_Budget", "USD_Worldwide_Gross", "USD_Domestic_Gross"]  
  
print(data[columns_to_analyze].describe())
```

Descriptive analysis of the dataframe:

	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
count	5,391.00	5,391.00	5,391.00
mean	31,113,737.58	88,855,421.96	41,235,519.44
std	40,523,796.88	168,457,757.00	66,029,346.27
min	1,100.00	0.00	0.00
25%	5,000,000.00	3,865,206.00	1,330,901.50
50%	17,000,000.00	27,450,453.00	17,192,205.00
75%	40,000,000.00	96,454,455.00	52,343,687.00
max	425,000,000.00	2,783,918,982.00	936,662,225.00

Mean budget: \$ 31,113,737.57837136

Mean worldwide gross revenue: \$ 88,855,421.96271564

Minimum revenue: \$ 0

The results show that the average production budget is approximately \$31.1 million, while the global average gross is approximately \$88.9 million, suggesting that, on average, films earn nearly three times their production costs.

At first glance, this seems encouraging and suggests that many films are profitable. However, the quartile values indicate a more nuanced picture.

The bottom 25% of films have an average budget of about \$5 million, but their global box office incomes total only about \$3.9 million, which means that a significant number of these films actually received losses.

Furthermore the dataset includes extremely high-budget productions, with the highest production budget reaching \$425 million and a maximum gigantic revenue that exceeds \$2.78 billion. This highlights the enormous scale of blockbuster films and the nature of the film industry, which is characterized by high risks and massive profits, with just a few major hits capable of generating extraordinary returns compared to an average film.

The \$0 values at revenue columns represents probably the films that were never released or had extremely limited distribution.



Domestic revenue represents box office income from the United States and Canada, while worldwide revenue includes both domestic and international incomes.

The average domestic revenue is approximately \$41 million, whereas the average worldwide revenue reaches about \$88 million.

This means that the United States and Canada alone account for nearly half of total global box office income, highlighting just how influential the

North American market is in determining a film's financial success — an astonishing share of the global movie industry.

Identifying Precise and Distinct Values in the Dataset

To better understand the scale of the dataset, we start by identifying movies with extreme production budgets. Using simple boolean filtering in pandas, we retrieve the entries corresponding to the lowest and highest recorded budgets.

```
# Finding particular entries
movie_lowest_budget = data.USD_Production_Budget == 1100.00
movie_highest_budget = data.USD_Production_Budget == 425000000.00

print("\nThe movie with the lowest Budget:")
print(data[movie_lowest_budget])
print("\nThe movie with the highest Budget:")
print(data[movie_highest_budget])
```

The movie with the lowest Budget:

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
5391	2005-05-08	My Date With Drew	1100	181041	181041

The movie with the highest Budget:

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
1	2009-12-18	Avatar	425000000	2783918982	760507625

This brief check reveals a striking contrast: **My Date with Drew** was produced on an extremely low budget of just \$1,100, while **Avatar** had the highest production budget at \$425 million. The enormous gap between these two films illustrates the wide range of film production scales—from independent passion projects to massive blockbuster productions.

Interestingly, despite its tiny budget, **My Date With Drew** grossed over \$181,000, representing an exceptionally high return on investment and serving as an excellent example of how even small independent productions can achieve considerable financial success.

Cleaning the Dataset: Investigating Missing Revenue Values

Before proceeding with a more in-depth analysis, it is important to review and clean up potentially misleading entries in the dataset. During our initial review, we noticed several films where the columns for domestic and worldwide box office revenue show a value of zero. Using simple Pandas filters, we isolate these cases to better understand whether they represent actual results, missing data, or films that were not released nationwide.

Movies With No Domestic (USA) Revenue

We first filter movies that reported zero domestic gross and sort them by production budget to inspect the most impactful cases.

This helps us quickly identify high-budget films that appear to have earned nothing in the United States.

```
non_profit_movies_usa = data[data.USD_Domestic_Gross == 0]
non_profit_movies_usa = non_profit_movies_usa.sort_values('USD_Production_Budget')

print("\nShape of the dataframe:")
print(non_profit_movies_usa.shape)

print("\nFirst five entries:")
print(non_profit_movies_usa.head())
```

Shape of the DataFrame:

```
(511, 6)
```

First five entries:

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
96	2020-12-31	Singularity	175000000	0	0
126	2018-12-18	Aquaman	160000000	0	0
321	2018-09-03	A Wrinkle in Time	103000000	0	0
366	2018-10-08	Amusement Park	100000000	0	0
566	2012-12-31	Astérix et Obélix: Au service de Sa Majesté	77600000	61319383	0

The result shows 511 movies with no recorded domestic revenue, including some surprisingly large-budget productions.

Movies With No Worldwide Revenue

Next, we'll filter out films with a global box office gross of zero, sort the list again, and select a few entries to examine more closely.

Using `.sample()` instead of `.head()` gives us a more representative snapshot of these entries.

```
non_profit_movies_worldwide = data[data.USD_Worldwide_Gross == 0]

non_profit_movies_worldwide =
    non_profit_movies_worldwide.sort_values('USD_Production_Budget')

print("\nShape of the dataframe:")
print(non_profit_movies_worldwide.shape)

print("\nRandom five entries:")
print(non_profit_movies_worldwide.sample(5))
```

Shape of the dataframe:

```
(354, 6)
```

Random five entries (we used `.sample()` here):

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
5308	2013-12-31	Echo Dr.	100000	0	0
5367	2015-04-21	The Front Man	20000	0	0
4523	1987-06-19	The Brave Little Toaster	2300000	0	0
5346	2013-10-25	Her Cry: La Llorona Investigation	35000	0	0
3487	2012-04-13	Detention	10000000	0	0

Here we find 354 movies with no recorded worldwide revenue.

Some of these are likely unreleased titles, very small independent productions, or cases where box office information was never properly tracked.

Interestingly, there are 511 movies with no domestic revenue, but only 354 with no worldwide revenue.

This difference indicates that a significant number of films earned money internationally while never being screened in the United States.

In fact, some movies generated several million dollars worldwide despite having zero domestic gross, highlighting how regional releases and distribution strategies can strongly influence box office performance.

For example, we see that the Italian movie “*Don Gato, el inicio de la pandilla*” (Top Cat Begins) made about \$4.5 million dollars in the box office, but nothing in the United States. Perhaps they should have screened it there too, considering it cost \$80 million to make.



These observations also remind us that zero values do not always mean failure — sometimes they simply reflect limited release markets or incomplete reporting, which is important to consider before performing profit-based analysis.

Filtering on multi conditions:

To better understand global distribution patterns, we will now focus on films that generated revenue outside North America but did not record any gross revenue in their domestic markets (the U.S. and Canada). These cases are particularly interesting because they highlight films that achieved international success without relying on the traditional Hollywood market.

To identify these entries, we apply multiple filtering conditions using pandas' query() function. This allows us to select films with zero domestic revenue but positive global revenue, effectively filtering out titles that were released exclusively outside the U.S. market.

```
international_releases = data.query('USD_Domestic_Gross == 0 and  
USD_Worldwide_Gross != 0')
```

After filtering, we sort the results by worldwide gross in descending order to examine the most successful films that never earned money domestically.

```
international_movies = international_movies.sort_values('USD_Worldwide_Gross',  
ascending=False)  
  
print("\nShape of the dataframe:")  
print(international_movies.shape)
```

```
print("\nThe 10 highest-grossing films released exclusively outside the US:")
print(international_movies.head(11))
```

Shape of the DataFrame:

```
(157, 6)
```

The 10 highest-grossing films released exclusively outside the US:

Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
2006-12-31	Les Bronzés 3: amis pour la vie	42000000	83833602	0
2015-08-05	La famille Bélier	11000000	74056184	0
2010-12-31	Sammy's avonturen: De geheime doorgang	25000000	71594792	0
2010-11-05	Haeundae	16000000	64733391	0
2012-12-31	Astérix et Obélix: Au service de Sa Majesté	77600000	61319383	0
2007-12-31	Taxi 4	17500000	60865364	0
2017-05-05	Chǎi dàn zhuānjiǎ	23000000	58807172	0
2010-12-31	Konferenz der Tiere	30000000	53048539	0
2017-08-25	Ballerina	30000000	48048527	0
2010-09-03	Hachiko: A Dog's Story	16000000	47707417	0
2009-12-31	Shi Yue Wei Cheng	23000000	44195779	0

The results highlight the strength of regional cinema and demonstrate that films can achieve impressive financial success even without a presence at the North American box office.

European productions dominate the list, particularly French comedies such as **Les Bronzés 3: amis pour la vie** and **La famille Bélier**, which performed exceptionally well thanks to the strong loyalty of domestic audiences and cultural familiarity.

Similarly, animated films such as **Sammy's avonturen: De geheime doorgang** and **Konferenz der Tiere** demonstrate that family-oriented content often performs well in European markets even without U.S. distribution.

A particularly interesting case is **Hachiko: A Dog's Story**. Although the film featured well-known Hollywood actors, it was not a box office hit in the U.S. but enjoyed great popularity internationally, especially in Japan. This example shows how emotional storytelling and cultural resonance can trump traditional Hollywood distribution strategies.

Overall, these results show that North America is not the only path to financial success. Strong regional appeal, local expertise, and targeted international releases can generate revenues in the tens of millions, underscoring the importance of global markets in the modern film industry.

Removing Problematic Entries Before Profit Analysis

Before analyzing profitability, we need to remove entries that would skew the results. Some films in the dataset have no revenue, not because they were flops, but because they have not yet been released or were never shown in theaters.

If we included these titles, they would be incorrectly classified as financial flops; therefore, we identify and remove them before proceeding.

Films Not Released Yet at Data Collection Time

The dataset was compiled on May 1, 2018, which means that films released in theaters after that date naturally have no box office revenue. To avoid misleading conclusions, we apply a time filter to exclude films that had not yet been released in theaters by that date.

Defining the Data Collection Date:

To correctly identify films that have not yet been released, we first use the date of the data query as a timestamp. This allows us to compare the theatrical release dates with the time the data was collected.

```
# First, create the time stamp:  
data_retrieving_date = pd.Timestamp('2018-5-1')
```

Filtering and Sorting the DataFrame:

Using the timestamp, we filter out films released after the data collection date and sort them by production budget. This allows us to quickly identify the most influential films that have not yet been released, which would otherwise skew the profit analysis.

```
no_revenue_data_movies = data[data.Release_Date >= data_retrieving_date]  
  
no_revenue_data_movies.sort_values("USD_Production_Budget", ascending=False)
```

Displaying:

```
print("\nMovies that haven't been released yet:")  
print(no_revenue_data_movies.shape)  
print(no_revenue_data_movies.head())
```

Movies that haven't been released yet::

(7, 6)

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
96	2020-12-31	Singularity	175000000	0	0
126	2018-12-18	Aquaman	160000000	0	0
321	2018-09-03	A Wrinkle in Time	103000000	0	0
366	2018-10-08	Amusement Park	100000000	0	0
1119	2020-12-31	Hannibal the Conqueror	50000000	0	0

This filtering reveals 7 unreleased films, including large-budget titles such as Singularity and Aquaman.

Their zero revenue values are therefore not meaningful, as these films simply had not been released at the time the dataset was collected.

Films That Never Had a Theatrical Box Office Release

Some films were scheduled before the date of the data entry but have no reported global box office revenue.

These titles were either canceled, postponed indefinitely, released directly on streaming platforms, or were unable to secure a theatrical release.

Building DataFrames using Multiple Conditions Filtering:

```
non_released_movies = data.query('USD_Worldwide_Gross == 0 and Release_Date < @data_retrieving_date')
non_released_movies = non_released_movies.sort_values("USD_Production_Budget",
ascending=False)
```

Displaying:

```
print("\nMovies that were not released due to the issues:")
print("\nShape of the dataframe:")
print(non_released_movies.shape)
print("\nSome of these movies")
print(non_released_movies.head())
```


Movies that were not released due to the Production issues:

(347, 6)

Rank	Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross
880	2015-11-12	The Ridiculous 6	60000000	0	0
1435	2015-12-31	The Crow (Reboot)	40000000	0	0
1631	2008-12-31	Black Water Transit	35000000	0	0
1656	2015-10-30	Freaks of Nature	33000000	0	0
1690	2004-12-31	George and the Dragon	32000000	0	0

That comes to a total of 347 films that were produced but never generated any box office revenue.

Some of them had a complicated release process—for example, “The Ridiculous 6” (2015) couldn’t be shown in theaters at all, but was able to premiere on Netflix, where the film reportedly attracted a huge audience despite the lack of traditional box office figures.

Other projects, such as The Crow (2015), faced production setbacks and were eventually shelved, while films like Black Water Transit struggled with distribution issues and never received a wide theatrical release.

These cases demonstrate that a lack of revenue does not always indicate failure, but rather points to alternative sales strategies or unresolved production issues.

Since these items cannot be appropriately evaluated in a profit-oriented analysis, excluding them ensures a more accurate financial comparison.

Eliminating Non-Representative Entries

After identifying both unreleased films and titles that were not released in theaters, we remove them using the `.drop()` function.

By passing the index of the filtered DataFrames, we ensure that exactly the right rows are removed from the original dataset without affecting other entries.

In this step we are going to create a new, cleaned dataset that contains only movies with meaningful revenue data suitable for financial analysis.

```
# We are going to remove them from the dataframe:
clean_data = data.drop(no_revenue_data_movies.index)
clean_data = clean_data.drop(non_released_movies.index)
```

```
remaining_movies = clean_data.shape[0]
print(f"The clean DataFrame now contains {remaining_movies} movies that are ready
for the analysis.")
```

The clean DataFrame now contains 5037 movies that are ready for the analysis.

After removing these problematic entries, the dataset now contains 5037 movies.

With unreleased titles and non-theatrical releases excluded, the data is now more consistent and reliable, allowing us to perform profit and revenue analysis without misleading zero-value distortions.

Profit Analysis of Movies

To evaluate financial performance, we calculate each film's profit by subtracting the production budget from the worldwide gross revenue. This allows us to look beyond mere revenue and better understand which films were actually financially successful.

Creating new Feature in the DataFrame based on current values:

We are going to create a new column for the Profits of movies. This transformation enables direct comparisons between movies and allows us to perform statistical analysis on profitability.

Since worldwide gross already includes both domestic and international earnings, profit can be computed up directly from the difference between total revenue and production cost.

```
data['Profit'] = data.USD_Worldwide_Gross - data.USD_Production_Budget
```

Descriptive Statistics for Budget and Profit:

After computing profit, we examine summary statistics to understand the overall distribution of budgets and financial outcomes across the dataset.

Now let's take a look at the overview of the financial data of movies

```
print(data[['USD_Production_Budget', 'Profit']].describe())
```

	USD_Production_Budget	Profit
count	5,037.00	5,037.00
mean	32,894,194.90	62,307,352.63
std	41,109,292.86	144,669,389.51
min	1,100.00	-110,450,242.00
25%	6,000,000.00	-1,923,346.00
50%	19,000,000.00	11,446,802.00
75%	42,000,000.00	66,511,453.00
max	425,000,000.00	2,358,918,982.00

The descriptive statistics reveal several interesting patterns about movie profitability.

The average production budget in the dataset is approximately \$32.9 million, suggesting that most films fall into the mid-range budget category. The average profit, on the other hand, is approximately \$62.3 million and indicates a profit on average.

However, the median profit is significantly lower, at around \$11.4 million, suggesting that a small number of extremely successful films are driving the average up considerably. This large difference between the mean and the median indicates a situation where a handful of popular movies generate enormous profits, while many films achieve only modest results.

Looking at the minimum figures, the largest loss amounts to approximately - \$110 million, showing that big-budget failures can lead to substantial financial losses. At the other end of the spectrum, the maximum profit exceeds \$2.35 billion, underscoring the immense earning potential of films that have been a global success.

The quartile values provide additional insight into profitability. The bottom 25% of films have profits below approximately -\$1.9 million, meaning that at least a quarter of the movies in the dataset actually lost money.

The positive median profit suggests that slightly more than half of the films were profitable, though many generated only modest returns. The top 25% of films generated a profit of more than \$66.5 million, indicating that major financial success is concentrated in a relatively small group of films.

Overall, these figures show that while the film industry can be extremely profitable, it is also fragile with high risks and highly inconsistent: While a few major hits generate enormous profits, a significant number of films struggle to break even or even incur losses.

Measuring the Share of Unprofitable Movies

To better understand the risks associated with film production, we are going to determine how many films in the dataset failed to turn a profit.

By filtering out films with negative profit figures and calculating their proportion, we can estimate how common financial losses are in the industry.

```
unprofitable_films = clean_data[clean_data.Profit < 0]

pct_loss_movies = (len(unprofitable_films) / len(clean_data))*100

print(f"% {pct_loss_movies}")
```

```
% 32.876
```

The results show that approximately 32.8% of films did not make money at the box office, meaning that nearly one third of the films resulted in a financial loss.

This highlights the enormous risk involved in filmmaking, where profitability is by no means guaranteed, even with substantial budgets and marketing efforts.

It also confirms the earlier observation that the industry is largely sustained by a small number of extremely successful films that compensate for many underperforming productions.

Moreover, since this calculation only considers production budgets and excludes additional expenses such as **marketing, distribution, and promotional campaigns**, the actual financial losses are likely to be even larger, indicating that a significant portion of films may perform far worse financially than this analysis alone suggests.

Profit Margin Analysis: Returns Relative to Budget

Now that we've examined absolute profits, let's turn our attention to profit margins, which measure how efficiently films have generated returns according to their production budgets.

This helps us identify films that have performed exceptionally well despite limited resources, rather than simply highlighting high-budget blockbusters.

By calculating profit as a percentage of production costs, we can compare films of very different scales on a uniform basis.

```
clean_data['Profit_Margin'] =  
    (clean_data.Profit / clean_data.USD_Production_Budget) * 100  
  
clean_data = clean_data.sort_values('Profit_Margin', ascending=False)  
  
print(clean_data.head(9))
```

Top Movies by Profit Margin:

Release_Date	Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross	Profit	Profit_Margin
1972-06-30	Deep Throat	25000	45000000	45000000	44975000	179,900.00
1980-03-21	Mad Max	200000	99750000	8750000	99550000	49,775.00
2009-09-25	Paranormal Activity	450000	194183034	107918810	193733034	43,051.79
2015-10-07	The Gallows	100000	41753881	22764410	41653881	41,653.88
1999-07-14	The Blair Witch Project	600000	248300000	140539099	247700000	41,283.33
2004-07-05	Super Size Me	65000	22233808	11529368	22168808	34,105.86
1942-08-13	Bambi	858000	268000000	102797000	267142000	31,135.43
1993-02-26	El Mariachi	7000	2041928	2040920	2034928	29,070.40
1968-01-10	Night of the Living Dead	114000	30000000	12000000	29886000	26,215.79

The top results are dominated by low-budget films that achieved extraordinary financial success.

Movies such as Mad Max, Paranormal Activity, The Blair Witch Project, and The Gallows demonstrate how modest production costs combined with strong audience interest can generate exceptional returns. Classic examples like Night of the Living Dead and El Mariachi also illustrate how independent productions can become cultural landmarks while delivering enormous financial efficiency.

The shared characteristics of the movies demonstrate how limited budgets and innovative storytelling can create a powerful impact without requiring massive investments.

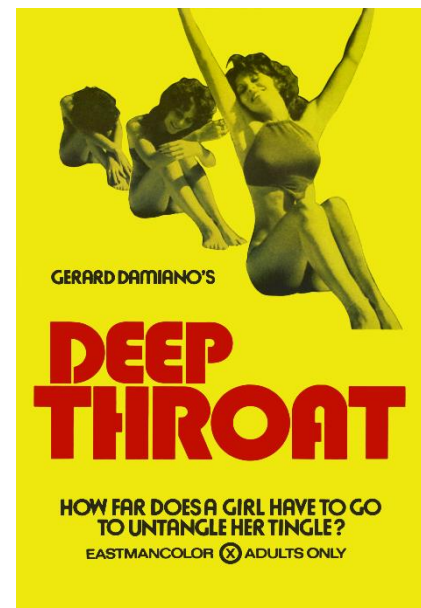
However, **Deep Throat (1972)** stands out as a clear anomaly.

While the other films in the list already boast exceptional profit margins ranging from 30,000% to 50,000%, this film achieves a **profit margin of nearly 180,000%**, far surpassing even the most successful independent films.

The financial history of “Deep Throat” is controversial; although the records show earnings of around **\$45 million**, some historical estimates put the figure as high as **\$600 million**.

These figures are disputed, as the film was released during the “porn chic” era of the 1970s, when reporting on the box office earnings of adult films was inconsistent and often influenced by theaters under **Mafia control**.

As a result, historians generally agree that while the film was undoubtedly extremely profitable, the figures regarding its highest box office earnings were likely exaggerated, meaning that the calculated profit margin should be viewed as an outlier and interpreted with caution.



Removing Extreme Outliers

To ensure the integrity of the analysis, we are going to exclude this movie from the dataset, as the reported financial figures are highly controversial and would disproportionately skew the distribution of profit margins.

Excluding this extreme outlier allows for a more realistic comparison among the remaining films.

```
deep_throat_1972 = clean_data[clean_data.Rank == 5356]

clean_data.drop(deep_throat_1972.index, inplace=True)
```

Descriptive analysis for Profit Margins:

```
print("\nProfit Margin Descriptive Analysis:")  
print(clean_data['Profit_Margin'].describe())
```

```
count    5,036.00  
mean       385.55  
std       1,889.50  
min       -100.00  
25%       -32.14  
50%        86.25  
75%       292.08  
max       49,775.00
```

```
Name: Profit_Margin, dtype: float64
```

After removing outliers, the descriptive statistics show an average profit margin of approximately 385%, while the median profit margin stands at 86%, suggesting that a typical profitable film nearly doubles its production budget on average.

The large gap between the maximum value (49,775%) and the upper quartile (292%) confirms that there are still independent hits with extremely high returns, but that these are rare, which supports our earlier observation that a small number of breakthrough hits account for a large portion of overall profitability.

Preparing for Visualization and Exploratory Analysis

Now that the dataset has been cleaned and the key financial metrics have been calculated, we can move on to the visualization phase.

The graphical analysis will allow us to better understand the relationships between budgets, revenue, profits, and the overall performance of the films.

In the following sections, we will examine key patterns and standout films using various visualizations, including bar charts, scatter plots, interactive graphics, and 3D representations. These visualizations help reveal trends, identify outliers, and provide intuitive insights into how films perform across different budgets and financial outcomes.

Movies with the Largest Production Budgets

To understand the scale of investment in major film productions, we are going to take a look at the films with the highest production budgets in cinema history. High-budget films are often ambitious projects that feature state-of-the-art special effects, a large cast, and high marketing expectations.

Visualizing these movies helps highlight how much studios are willing to invest in blockbuster productions and sets the stage for comparing budget size with financial performance.

Selecting the Top 15 Highest-Budget Films

To focus on the most expensive productions, we sort the dataset by production budget in descending order and select the top 15 entries:

```
top_15_budget =  
    clean_data.sort_values('USD_Production_Budget', ascending=False)[:15]
```

Creating Bar Graph:

Code:

```
sns.set_theme(style="whitegrid", context="talk")  
fig, ax = plt.subplots(figsize=(16, 8), dpi=300)  
  
sns.barplot(  
    x=top_15_budget.Movie_Title,  
    y=top_15_budget.USD_Production_Budget,  
    hue=top_15_budget.Movie_Title,  
    palette="Set2", edgecolor=".2", width=0.75,  
    legend=False  
)  
  
plt.title('Top 15 Movies with the Highest Production Budgets', fontsize=22,  
fontweight='bold', pad=25, color='#222222')  
plt.xlabel('')  
plt.ylabel('Production Budget', fontsize=12, fontweight='bold', labelpad=10)  
plt.xticks(fontsize=12, fontweight='bold')  
plt.yticks(fontsize=12)  
  
# wrap labels  
labels = ax.get_xticklabels()  
new_labels = [textwrap.fill(label.get_text(), 10) for label in labels]  
ax.set_xticks(ax.get_xticks())  
ax.set_xticklabels(new_labels)
```

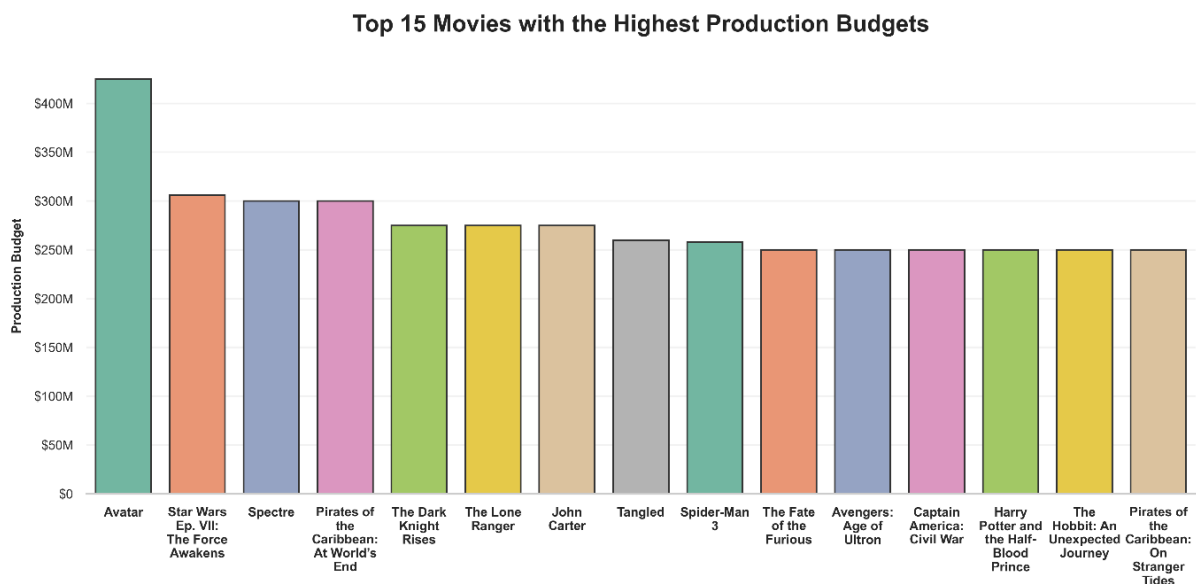


```
# formatting the axes for money-wise designing (Converts numbers to $M and $B)
def currency_formatter(x, pos):
    if x >= 1e9: return f'${x*1e-9:.1f}B'
    elif x >= 1e6: return f'${x*1e-6:.0f}M'
    elif x == 0: return '$0'
    return f'${x}'

ax.yaxis.set_major_formatter(FuncFormatter(currency_formatter))
ax.yaxis.grid(True, alpha=0.25)
ax.set_axisbelow(True)

sns.despine(left=True, top=True, right=True)
plt.tight_layout()

plt.show()
```



The chart shows that most of the highest-budget films cluster between \$200M and \$300M, indicating a common investment range for modern blockbuster productions.

Avatar clearly stands apart, exceeding \$400M, making it a significant outlier and underscoring the extraordinary scale of its production.

Another interesting observation is the dominance of franchise films, all of them are science-fiction movies such as Marvel, Star Wars, Harry Potter, Pirates of the Caribbean, which suggests that studios are more willing to spend massive budgets to established film universes with predictable global demand.

Movies with the Highest Profits

Profit is one of the most direct indicators of a film's financial success and reflects the revenue remaining after production costs have been deducted. While box office payoffs alone can be misleading, profit reveals which films have actually generated high financial returns.

Here we are going to create a bar graph which shows the films that have generated the highest absolute profits.

Preparing the Dataset

To identify the most profitable movies, the dataset was sorted based on the calculated profit values in descending order. The top 15 entries were then selected to highlight the films that generated the largest financial gains.

```
# Top 15 Movies with the highest profits:
top_15_profit = clean_data.sort_values('Profit', ascending=False)[:15]
```

Creating the Bar Graph

```
sns.set_theme(style="whitegrid", context="talk")
fig, ax = plt.subplots(figsize=(16, 8), dpi=300)

sns.barplot(
    x=top_15_profit.Movie_Title,
    y=top_15_profit.Profit,
    hue=top_15_profit.Movie_Title,
    palette="muted",
    edgecolor=".2",
    width=0.75,
    legend=False
)

plt.title('Top 15 Movies with the Highest Box-Office Profit', fontsize=22,
fontweight='bold', pad=25, color='#222222')
plt.xlabel('')
plt.ylabel('Profit', fontsize=12, fontweight='bold', labelpad=10)
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12)

labels = ax.get_xticklabels()
new_labels = [textwrap.fill(label.get_text(), 10) for label in labels]
ax.set_xticks(ax.get_xticks())
```

```

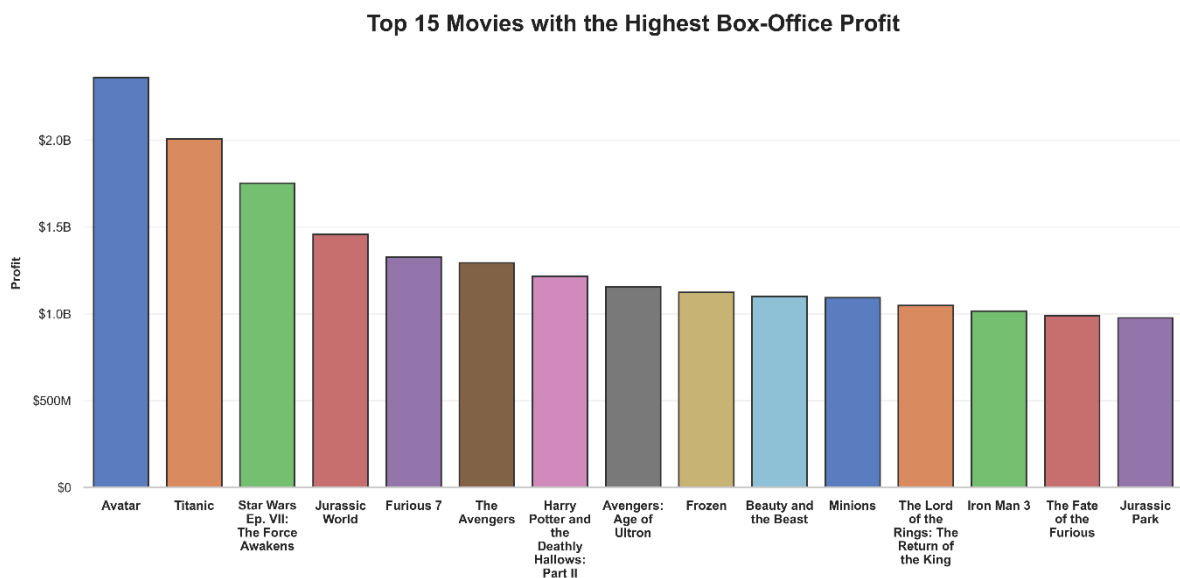
ax.set_xticklabels(new_labels)

# We are going to use currency_formatter that we created previously.
ax.yaxis.set_major_formatter(FuncFormatter(currency_formatter))
ax.yaxis.grid(True, alpha=0.25)
ax.set_axisbelow(True)

sns.despine(left=True, top=True, right=True)
plt.tight_layout()

plt.show()

```



This chart presents the top 15 movies by absolute box-office profit, with Avatar, Titanic, and Star Wars standing noticeably ahead of the rest. After these leaders, films such as Jurassic World, Furious 7, and The Avengers form a tighter group, where profits remain extremely high but differences between entries become smaller.

The list mainly consists of action-heavy franchise films, which typically benefit from massive budgets and an already established fan base that ensures strong revenues.

Interestingly, “Titanic” stands out as the only primarily drama movie among the other movies highlighting its extraordinary cultural significance and enduring appeal to audiences.

Furthermore, the presence of animated films such as “Frozen,” “Minions,” and “Beauty and the Beast” suggests that family-oriented films can achieve remarkable profitability.

Top 20 Movies with the Highest Profit Margins

The profit margin indicates how efficiently a film converts its production budget into profit, making it a useful metric for comparing the financial performance of films of varying scales.

Unlike absolute profit, this approach highlights the films that have generated the highest return on investment relative to their budgets.

Preparing the Dataset — Filtering Low-Budget Outliers

Extremely high profit margins are often associated with very low-budget films, which can skew comparisons with mainstream productions.

Based on the descriptive statistics, the average production budget in the dataset is approximately \$30 million, which serves as an appropriate threshold for distinguishing small-scale productions from widely distributed studio films.

To ensure meaningful and comparable results, films with production budgets under \$30 million were excluded from the analysis.

```
mean_budget_at_all_movies = 30000000

movies_over_avg = clean_data.query('USD_Production_Budget >=
@mean_budget_at_all_movies')

print(movies_over_avg.shape)
```

After applying this filter, 1,851 films remain in the dataset.

These films are medium- to high-budget productions, allowing their profit margins to be compared without being skewed by outliers with micro-budgets.

The next step is to identify the most efficient films among them based on their profit margins as a percentage.

Selecting the Top Profit Margin Movie

The filtered dataset was sorted by profit margin to identify the top 20 movies that generated the highest return relative to their production budgets.

```
movies_over_avg = movies_over_avg.sort_values('Profit_Margin',
ascending=False)[:20]
movies_over_avg = movies_over_avg.sort_values('Profit_Margin', ascending=True) #
it should be descending for horizontal graph
```

Creating the Interactive Horizontal Bar Chart

To better illustrate the films with the highest profit margins, an interactive horizontal bar chart was created. Because the chart is interactive, additional contextual information appears when the mouse pointer hovers over the individual bars, allowing the viewer to explore further details without the visualization feeling cluttered.

Preparing Additional Display Information

To complement the interactive chart, additional information such as release year, production budget, and total gross was compiled. The release dates were first converted into release years to present a more concise and meaningful time reference in the visualization.

The extracted release years were then combined with production budgets and profit figures to create structured display elements.

These elements are used in the interactive chart to display additional details when the user hovers over a film with the mouse.

[illegible]

code for the graph:

```
bar = px.bar(
    movies_over_avg_profit,
    x='Profit_Margin',
    y='Movie_Title',
    orientation='h',
    color='Profit_Margin',
    color_continuous_scale='phase',
    title='Top 20 Movies with the Highest Profit Margins'
)

# Options for styling, text formatting, and hover template
bar.update_traces(
    text=movies_over_avg_profit['Profit_Margin'], # Display the profit margins
    texttemplate='<b>{%text:,.0f}%</b>', # Formatting profit margins to percentages.
    textposition='outside', textfont_size=12, marker_line_width=0,
    customdata=display_components,
    hovertemplate=(
        "<b>{%y}</b><br>" +
        "Release Year: {%customdata[0]}<br>" +
        "Budget: ${{customdata[1]:,.0f}}<br>" +
        "Profit: ${{customdata[2]:,.0f}} " +
        "<extra></extra>"
    )
)

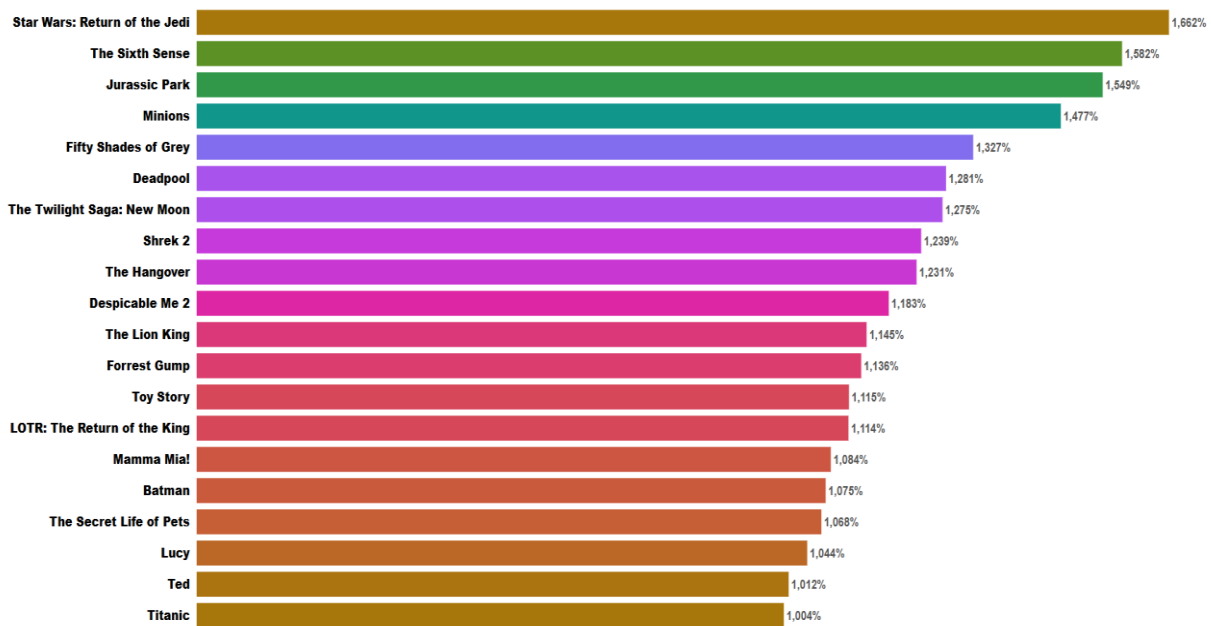
bar.update_layout(
    template='plotly_white',
    title=dict(text='<b>Top 20 Movies with the Highest Profit Margins</b>',
               font=dict(size=22, family='Arial, sans-serif', color='#2C3E50')),
    x=0.02),
    xaxis_title='', yaxis_title='', # No title, they are obvious
    font=dict(family='Arial, sans-serif', size=13, color='#555555'),
    showlegend=False, # No Legend
    coloraxis_showscale=False, # No coloraxis
    xaxis=dict(showgrid=False, showticklabels=False, zeroline=False),
    yaxis=dict(showgrid=False, ticklabelposition="outside"),
    margin=dict(l=10, r=70, t=70, b=20),
)

bar.write_html("interactive_graphs/movies/top_20_profit_margin_movies.html")

bar.show()
```

Graph:

Top 20 Movies with the Highest Profit Margins



Interactive Graph Link: [Top 20 Movies with the Highest Profit Margins](#)

This figure shows the films with the highest profit margins, illustrating how efficiently certain films were able to turn modest budgets into enormous profits. Movies such as “Star Wars: Return of the Jedi,” “The Sixth Sense,” and “Jurassic Park” stand out at the top; each of them generated more than fifteen times its production budget, suggesting exceptional economic efficiency rather than merely large investments.

Several of these films achieved high profit margins by combining controlled budgets with broad appeal to audiences; for example, “The Sixth Sense” relied heavily on storytelling, while “Deadpool” and “The Hangover” benefited from distinctive marketing and strong audience-driven popularity despite their moderate budgets.

Animated films such as “Minions,” “Despicable Me 2,” and “The Lion King” also perform remarkably well, suggesting that family-oriented content can generate repeat viewings and strong global demand relative to production costs.

Hovering over the bars also reveals that many of these films were produced with budgets ranging from approximately \$30 million to \$80 million, which supports the notion that moderate investments, when combined with strong concepts, franchise recognition, or cultural relevance, can yield exceptionally high returns.

Top Movies with the Largest Box Office Losses

While some films generate extraordinary profits, others fail to recoup their production costs and result in significant financial losses.

In this part, we are going to analyze these deficits to shed more light on the risks associated with big-budget filmmaking and to illustrate that even highly anticipated productions can fall short of expectations.

This analysis focuses on identifying the films with the highest absolute losses in the dataset.

Extracting Movies with the Biggest Deficits

To identify the largest financial failures, we are going to select the lowest-performing movies and convert their losses into absolute values to simplify visualization and comparison.

This approach ensures that the chart focuses on the magnitude of financial losses, making it easier to compare the scale of underperformance across films.

```
movies_biggest_loss = clean_data.sort_values('Profit', ascending=True)[:20]

# Creating absolut loses column to design the graph more easily:
movies_biggest_loss['Absolute_Loss'] = movies_biggest_loss['Profit'].abs()
movies_biggest_loss.sort_values(by='Absolute_Loss', ascending=False)
```

Preparing Hover Display Information

Additional contextual information such as release year, production budget, worldwide gross revenue, total loss, and profit margin was prepared to be displayed when hovering over each bar in the interactive visualization to provide deeper insight.

[illegible]

Code for the graph:

```
bar = px.bar(
    movies_biggest_loss,
    x='Absolute_Loss',
    y='Movie_Title',
    orientation='h',
    color='Absolute_Loss',
    color_continuous_scale=red_colors
)

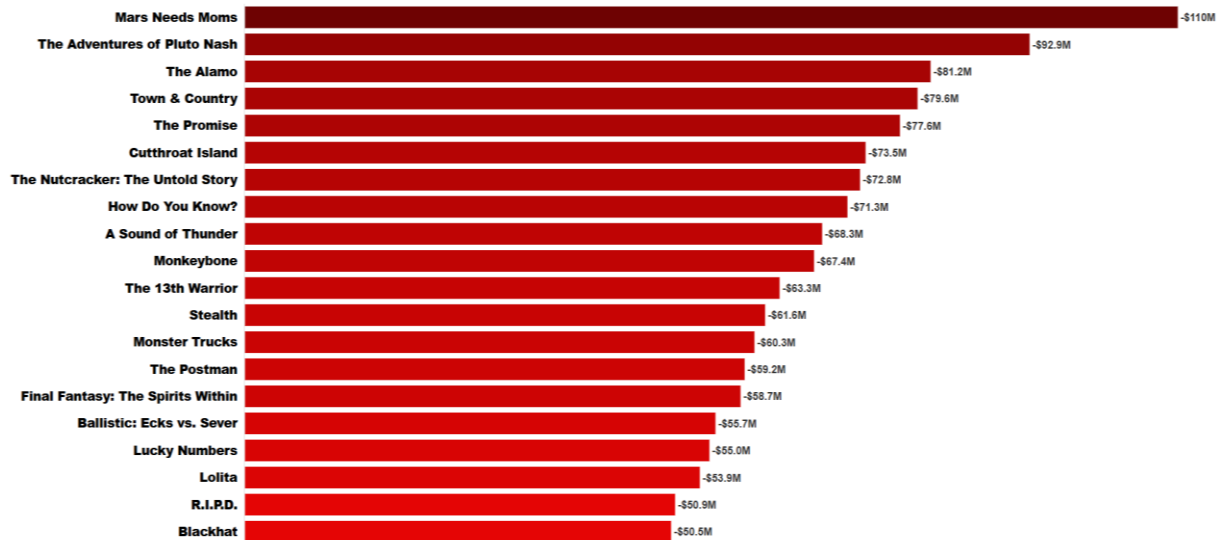
bar.update_traces(
    text=movies_biggest_loss['Profit'],
    texttemplate='<b>{%text:$,.3s}</b>',
    textposition='outside',
    textfont=dict(size=12, family='Inter, Arial, sans-serif', color='#333333'),
    marker_line_width=0,
    customdata=display_components,
    hovertemplate=(
        "<b style='font-size:15px; color:#1A1A1A;'>{%y}</b><br><br>" +
        "<span style='color:#555555;'>Release Year:</span> <b style='color:#1A1A1A;'>{%customdata[0]}</b><br>" +
        "<span style='color:#555555;'>Budget:</span> <b style='color:#1A1A1A;'>{%customdata[1]:$,.0f}</b><br>" +
        "<span style='color:#555555;'>Worldwide Gross:</span> <b style='color:#1A1A1A;'>{%customdata[2]:$,.0f}</b><br>" +
        "<span style='color:#D32F2F;'>Loss:</span> <b style='color:#D32F2F;'>{%customdata[3]:$,.0f}</b><br>" +
        "<span style='color:#555555;'>Loss Rate:</span> <b style='color:#1A1A1A;'>{%customdata[4]:.1f}%</b><br>" +
        "<extra></extra>"
    )
)

bar.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Top 20 Movies with the Biggest Losses</b>', x=0.02, y=0.95,
        font=dict(size=22, family='Inter, Arial, sans-serif', color='#c60404')),
    xaxis_title='', yaxis_title='',
    font=dict(family='Inter, Arial, sans-serif', size=13, color='#555555'),
    showlegend=False,
    coloraxis_showscale=False,
    xaxis=dict(showgrid=False, showticklabels=False, zeroline=False,
        tickfont=dict(size=14, family="Arial Black", color='black')),
    yaxis=dict(showgrid=False, ticklabelposition="outside", ticklabelstandoff=5,
        autorange="reversed", linecolor='#E0E0E0', linewidth=1,
        tickfont=dict(size=14, family="Arial Black", color='black')),
    margin=dict(l=10, r=90, t=80, b=20),
    hoverlabel=dict(
        bgcolor="white",
        bordercolor="#E0E0E0",
        font_size=13,
        font_family="Inter, Arial, sans-serif")
)

bar.write_html("interactive_graphs/movies/top_20_losses_movies.html")
```

```
bar.show()
```

Top 20 Movies with the Biggest Losses



Interactive Graph Link: [Top 20 Movies with the Biggest Losses](#)

This figure shows the films that have incurred the highest absolute financial losses, with titles such as “Mars Needs Moms,” “The Adventures of Pluto Nash,” and “The Alamo” standing out as the biggest box office flops.

Most films on the list combine high production budgets with very low global box office revenues, resulting in losses ranging from over \$50 million to \$110 million, as confirmed by the detailed information that appears when we hover over the graph.

Interestingly, the graph illustrates two distinct types of failure: Some films, such as *Mars Needs Moms*, incurred significant losses primarily due to very high budgets, while others, such as *Lolita*, performed extremely poorly relative to their investments and lost nearly their entire budget.

This distinction suggests that both excessive spending and a weak audience response can independently lead to significant financial losses, even if one of the factors is only moderately pronounced.

Overall, the results underscore the high-risk nature of big-budget filmmaking, where unsuccessful releases can lead to substantial deficits despite significant production investments.

General Financial Performance of Movies

So far, we have been focused on individual outliers such as the highest budgets, the largest profits, the biggest losses, and the most efficient profit margins. These sections have highlighted standout films and demonstrated how certain productions significantly exceeded or fell short of expectations.

However, looking only at the top performers or the bottom performers does not provide a complete picture of the overall financial situation.

To better understand broader patterns and relationships, we are going to conduct a more general analysis.

Moving to a Holistic View

In this section we are going to examine how production budgets, worldwide gross revenue, and profits interact across the entire dataset. Rather than focusing on individual outliers, this approach allows us to identify trends, clusters, and general performance patterns in the film industry.

We are going to explore if higher budgets consistently lead to higher profits or films with moderate budgets can compete with big-budget productions.

Creating a 3D Financial Performance Scatter Plot

To illustrate these relationships, we are going to create a 3D scatter plot.

In this chart, the production budget is plotted on the x-axis and global gross revenue on the y-axis, with both the color and size of the data points reflecting profit.

This multidimensional visualization allows us to simultaneously examine investment volume, revenue trends, and financial performance.

```
sns.set_theme(style="whitegrid", rc={"axes": "#FBFFFC", "figure": "#ffffff"})
fig, ax = plt.subplots(figsize=(16, 8), dpi=300)

# we are going to map the column Profit to both color (hue) and size
scatter = sns.scatterplot(
    data=clean_data,
    x='USD_Production_Budget',
    y='USD_Worldwide_Gross',
```

```

    hue='Profit',
    size='Profit',
    sizes=(20, 300),          # arranging bubble sizes
    alpha=0.6,                # Transparency to handle the overlapping points
    palette="crest",          # Elegant blue-green modern palette
    edgecolor="white",        # Slight white border around bubbles for crispness
    linewidth=0.5,
    ax=ax
)

# We are going to use currency formatter that we created previously.
ax.xaxis.set_major_formatter(FuncFormatter(currency_formatter))
ax.yaxis.set_major_formatter(FuncFormatter(currency_formatter))

# setting limits just for the arrangement at the bottom of the graph
ax.set_ylim(-50000000, 300000000)
ax.set_xlim(-10000000, 450000000)

# Axis Labels
ax.set_xlabel('Production Budget', fontsize=14, fontweight='bold',
color='#333333', labelpad=12)
ax.set_ylabel('Worldwide Gross Revenue', fontsize=14, fontweight='bold',
color='#333333', labelpad=12)

# main title
fig.text(0.08, 0.96, 'Financial Performance of Movies', fontsize=18,
fontweight='bold', color='#2c3e50')

# subtitle
fig.text(0.08, 0.92, 'Comparing Production Budgets to Worldwide Gross and
Profitability', fontsize=12, color='#7f8c8d')

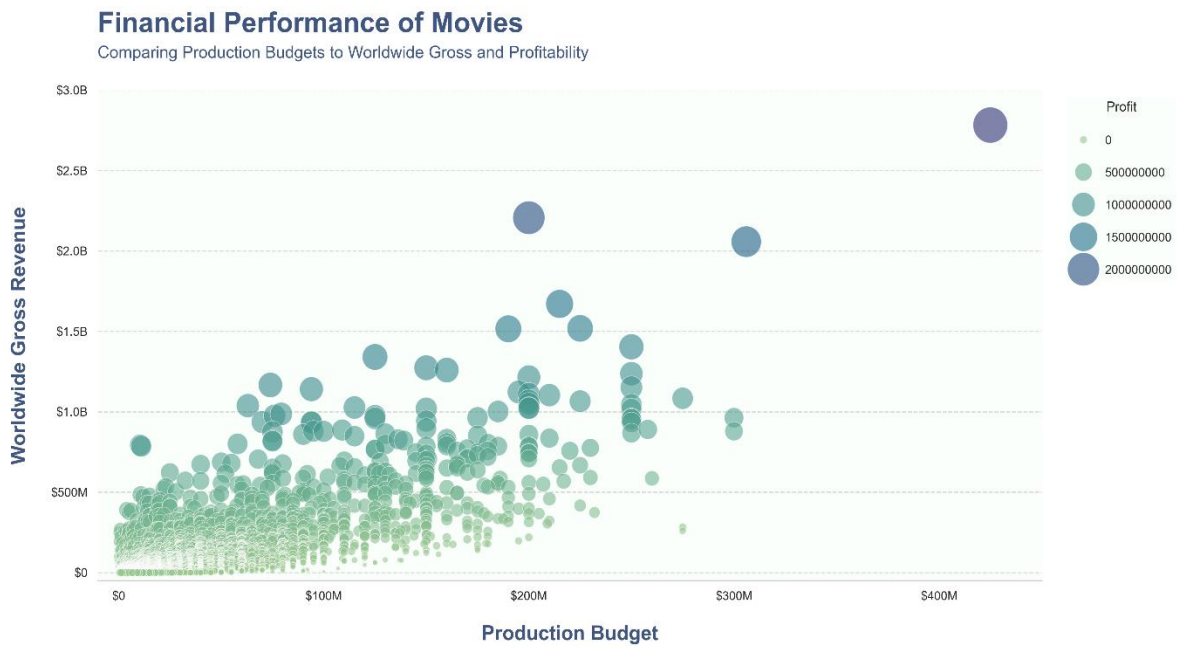
# arrangement for the Legend
legend = ax.legend(bbox_to_anchor=(1.02, 1), loc='upper left', frameon=True,
title="Profit", title_fontsize=12)
legend.get_frame().set_linewidth(0.0) # Remove border around Legend

# Cleaning up some borders and refining grid lines
sns.despine(left=True, bottom=False)
ax.grid(axis='y', linestyle='--', alpha=0.7) # Keeping horizontal gridlines
ax.grid(axis='x', visible=False)             # Removing vertical Lines

# adjusting layout so the Legend fits properly on the screen
plt.tight_layout(rect=[0, 0, 0.95, 0.9])

plt.show()

```



This scatter plot illustrates the relationship between production budgets and worldwide gross revenue, with bubble size and color indicating overall profit.

A general upward trend can be observed, suggesting higher budget films tend to generate higher revenues; however, the wide variation shows that budget alone does not guarantee financial success. Many mid-budget films generate box office earnings comparable to those of big-budget productions, which shows that a strong concept can make a bigger impact than mere financial resources.

The largest and darkest bubbles appear in the upper right quadrant and represent blockbuster films that combined high budgets with exceptional global box office incomes and ultimately generated the highest profits.

However, the dense cluster in the lower left corner shows that most films operate on modest budgets and generate moderate box office revenues, reflecting the typical financial scale of the industry.

Interestingly, some high-budget projects are falling short of expected revenue, highlighting the financial risk associated with expensive productions, where underperformance can quickly undermine profitability.

Overall, the analysis shows that while larger investments increase the potential for high returns, efficient budgets in the mid-range often lead to more consistent and less risky financial outcomes.

Evolution of Movie Production Budgets Over Time

Production budgets in the film industry have changed significantly over the years, due to technological advances, rising marketing costs, and growing audience expectations.

A chronological analysis of budgets helps to reveal whether films are becoming more expensive and how the level of investment correlates with financial success. From this perspective, it is also possible to observe during which periods high-budget film production gained prominence and whether larger investments consistently led to higher profits. By examining these trends, we can better understand the long-term financial development of the film industry.

Creating a 3D Scatter Plot of Budget Trends Over Time

To examine this relationship, we are going to generate a three-dimensional scatter plot, with the release date and the production budget on the axes, while profit is illustrated by both the color intensity and the size of the data points.

This visualization will allow us to simultaneously track how budgets change over time and how profitability varies at different investment levels.

```
sns.set_theme(style="whitegrid", font="sans-serif")
plt.figure(figsize=(16, 8), dpi=300)

ax = sns.scatterplot(
    data=clean_data,
    x='Release_Date',
    y='USD_Production_Budget',
    hue='Profit', size='Profit',
    sizes=(20, 700),      # a big size range for better contrast
    alpha=0.75,           # more transparency to see the overlapped bubbles
    palette="crest", edgcolor="white", linewidth=0.5
)

# we want to increase the values by 50 million dollars.
# For currency we will use the same currency formatter.
ax.yaxis.set_major_locator(ticker.MultipleLocator(50_000_000))
ax.yaxis.set_major_formatter(ticker.FuncFormatter(currency_formatter))

ax.set_title("Movie Budgets Over the Years", fontsize=25, fontweight='bold')
ax.set_xlabel("Release Year", fontsize=18, fontweight='bold', color='#405680')
ax.set_ylabel("Production Budget ($)", fontweight='bold', fontsize=18)
# arrangement for the Legend:
```

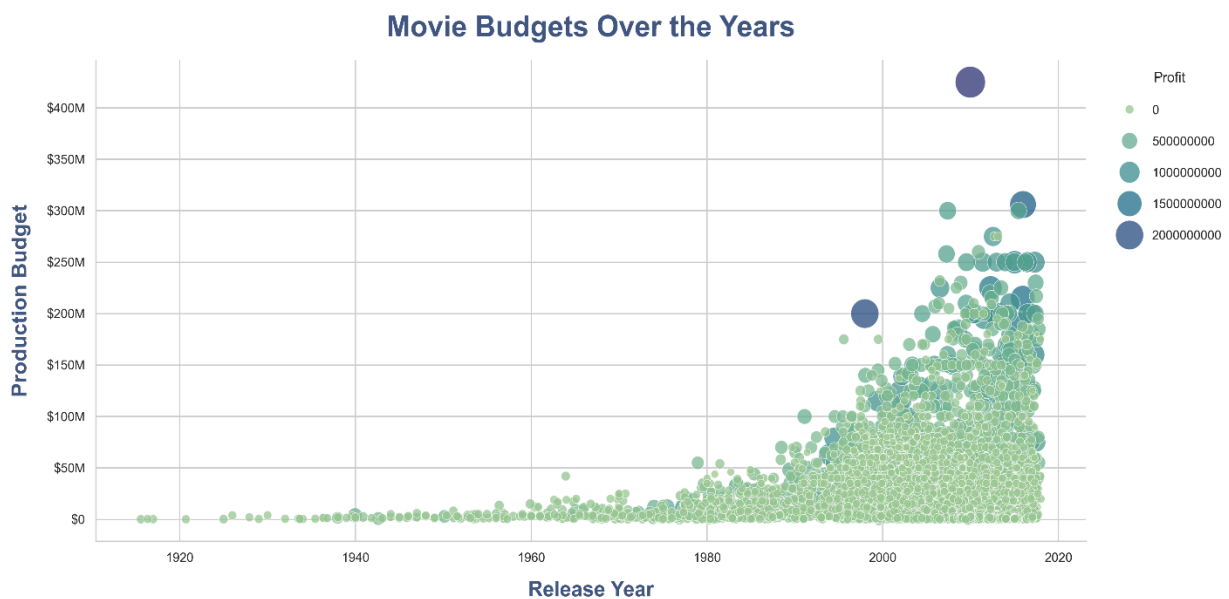
```

legend = ax.legend(bbox_to_anchor=(1.02, 1), loc='upper left', frameon=True,
title="Profit", title_fontsize=12, labelspace=1.5)
legend.get_frame().set_linewidth(0.0) # Remove border around Legend

# adjusting the layout
sns.despine(left=False, bottom=False)
plt.tight_layout(rect=[0, 0, 0.95, 0.9])

plt.show()

```



This graph shows a clear upward trend in film production budgets over time: relatively low and stable spending in earlier decades was followed by a sharp increase starting in the 1990s, which accelerated greatly after 2000.

The concentration of high-budget productions in recent years suggests that modern filmmaking is increasingly relying on large-scale productions, a trend likely attributable to advances in visual effects and global distribution.

The largest bubbles, which represent highly profitable films, are concentrated primarily in the realm of major productions from the 2000s and 2010s, suggesting that high investments are often accompanied by blockbuster-level returns.

However, the wide variation in profits in recent years also highlights the growing financial risk, as similarly high budgets can lead to very different profit shortfalls depending on audience response and market trends.

Identifying a Standout Movie from the Budget Trend

When looking at the chart, one bubble immediately catches the eye, a large, dark blue dot that appears before the year 2000. It represents the first film in the dataset with a production budget exceeding \$200 million, and its size and intense color suggest that it also generated substantial profits.

After this point, high-budget productions become far more common, suggesting that this film may have played a role in redefining expectations for blockbuster-scale investments. This makes it an interesting candidate for closer examination.

Locating the Movie in the Dataset

To identify this outlier, we can filter the dataset based on two conditions: a production budget of at least \$200 million and a release date prior to the year 2000. These criteria isolate the early, high-budget outlier visible in the visualization.

We are going to use `.query()` method to apply the conditions, and `.iloc[0]` to retrieve the corresponding movie details.

```
year_2000 = pd.Timestamp('2000-01-01')

revolutionary_movie = clean_data.query('USD_Production_Budget >= 200000000 and
Release_Date < @year_2000')

print("\nThe first film with a budget of over $200 million that went on to
generate enormous profits:")

print(revolutionary_movie.iloc[0])
```

The first film with a budget of over \$200 M and went on to generate enormous profit:

Rank	37
Release_Date	1997-12-19 00:00:00
Movie_Title	Titanic
USD_Production_Budget	200,000,000
USD_Worldwide_Gross	2,207,615,668
USD_Domestic_Gross	658,672,302
Profit	2,007,615,668
Profit_Margin	1,003.81
Name: 1034, dtype: object	



The result confirms that Titanic (1997) was the first film to exceed a production budget of \$200 million while also generating exceptionally high profits.

Its enormous financial success demonstrated that extremely high-budget productions could be commercially viable, which likely encouraged studios

to invest more aggressively in the production of large-scale blockbusters.

The significant increase in films with budgets exceeding \$200 million suggests that Titanic may have played a decisive role in changing industry expectations and ushering in the modern era of mega-budget cinema.

Segmenting Movies into Historical Periods

The film industry has evolved significantly over time, with production scale, distribution reach, and revenue potential changing considerably over the decades.

To better understand these structural changes, we are going to divide the dataset into two broad sections representing older and more modern eras of filmmaking.

This segmentation allows us to compare financial characteristics and explore how the economic conditions of the film industry have changed over time.

Creating a Decade Feature from Release Dates

To group films by historical period, we are going to convert release dates into decades. This conversion simplifies comparisons over time and makes it easier to analyze long-term trends rather than focusing on individual years. Each film will be assigned to the appropriate decade based on its release year.

```
release_date_indexes = pd.DatetimeIndex(clean_data.Release_Date)
years = release_date_indexes.year
decades = years//10*10
clean_data['Decade'] = decades
```

Defining the Boundary Between Old and Modern Films

To distinguish between older and more modern productions, we are going to select the year 1960 as a separation point.

The year 1960 and its era roughly mark the transition from early studio-era filmmaking to a more modern industry characterized by expanding international distribution, technological development, and increasing production budgets.

```
old_films = clean_data[clean_data.Decade <= 1960]

new_films = clean_data[clean_data.Decade > 1960]
```

Highest-Grossing Movies from the Pre-1960 Era

```
old_films.sort_values('USD_Worldwide_Gross', ascending=False)

print(old_films.head())
```

Movie_Title	USD_Production_Budget	USD_Worldwide_Gross	USD_Domestic_Gross	Profit	Profit_Margin	Decade
Gone with the Wind	3900000	390525192	198680470	386625192	9,913.47	1930
The Sound of Music	8200000	286214286	163214286	278014286	3,390.42	1960
Bambi	858000	268000000	102797000	267142000	31,135.43	1940
Cinderella	2900000	263591415	85000000	260691415	8,989.36	1950
Snow White and the 7 Dwarfs	1488000	184925485	184925485	183437485	12,327.79	1930

The results show that several classic films, such as Gone with the Wind, The Sound of Music, Bambi, and Snow White and the Seven Dwarfs, generated remarkable box office revenues despite relatively low production budgets.

These extremely high profit margins illustrate how early cult films benefited from long theatrical runs, limited competition, and a strong cultural impact, enabling them to achieve exceptional economic efficiency compared to modern productions.

In the next section we are going to examine the differences between old and new films in more clear and holistic way.

Comparing Budget–Revenue Trends Across Film Eras

To better understand the relationship between production budgets and global box office revenues, here we are going to generate two scatter plots with regression lines for both classical and modern films.

Unlike simple scatter plots, regression analyses help identify the overall trend and the strength of the relationship between investment and financial return. This approach makes it easier to compare whether higher budgets in different eras of filmmaking have led to higher revenues. Here we are going to also investigate how the ratio of worldwide incomes to production budgets has changed over time.

We are going to visualize the relationship between the production budget and global gross revenue with using Seaborn's `regplot()` function.

This function creates a scatter plot of the data points and simultaneously fits a linear regression line. Additionally, it also displays a confidence interval around the regression line, providing a visual indication of the variability and reliability of the trend.

Creating Scatter Plot with Linear Regression:

code:

```
sns.set_theme(style="whitegrid", rc={"axes.facecolor": "#fbfbfb", "grid.color": "#e9ecef"})
plt.figure(figsize=(16, 8), dpi=300)

ax = sns.regplot(
    data=old_films,
    x='USD_Production_Budget',
    y='USD_Worldwide_Gross',
    color="#2c3e50",
    line_kws={'color': '#DC143C', 'linewidth': 3 }, #regression line
    scatter_kws={
        'alpha': 0.6,          # transparency for the dots
        's': 90,               # marker sizes
        'edgecolor': 'white',  # a subtle border for the dots
    }
)

ax.xaxis.set_major_formatter(ticker.FuncFormatter(currency_formatter))
ax.yaxis.set_major_formatter(ticker.FuncFormatter(currency_formatter))

plt.suptitle('Movies before 1960', fontsize=18, color='#333333',)
```

```
plt.title('Production Budget vs. Worldwide Gross', fontsize=22, fontweight='bold',
pad=20, color='#DC143C', loc='left')
plt.xlabel('Production Budget', fontsize=14, fontweight='bold', color='#DC143C')
plt.ylabel('Worldwide Gross', fontsize=14, fontweight='bold', labelpad=12)

# these lines soften the tick labels
plt.xticks(fontsize=11, color='#666666')
plt.yticks(fontsize=11, color='#666666')

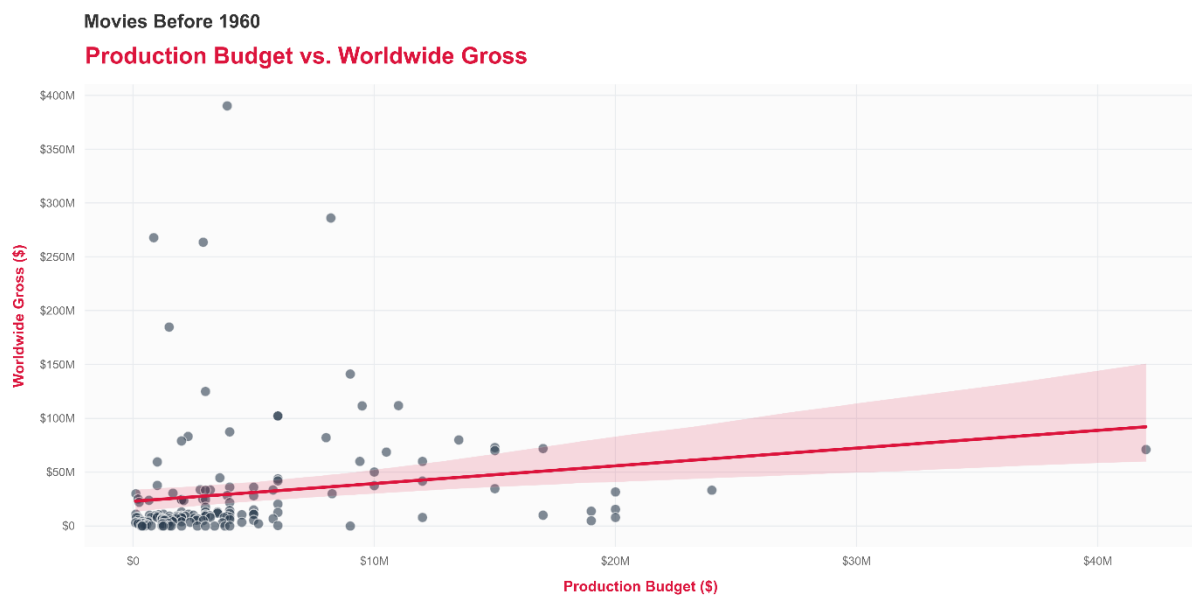
# Arrangements for the graph layout
sns.despine(left=True, bottom=True)
plt.tight_layout()

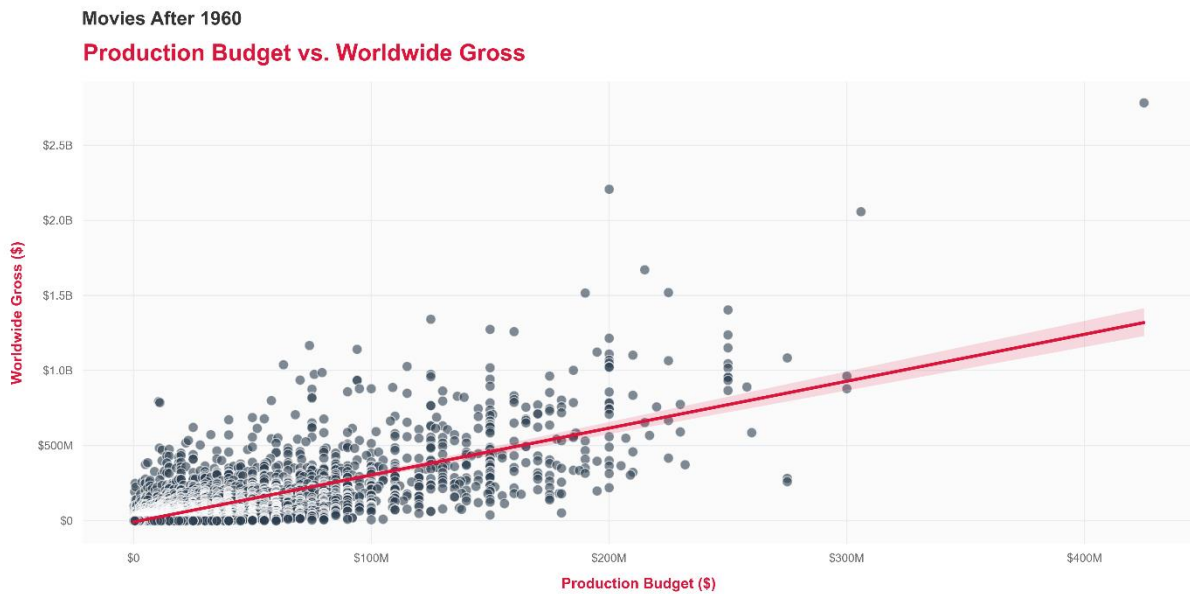
# Render the plot
plt.show()
```

Here, we are plotting for old movie data.

Same graph will also be applied to new film data with subtle changes at the code.

Graphs:





These regression plots reveal a clear structural difference in the budget–revenue relationship before and after 1960.

In the earlier period, the regression line shows only a slight upward trend, and the data points are widely scattered, suggesting that higher production budgets did not consistently lead to higher revenue. The wide confidence interval and the presence of several outliers suggest that a handful of exceptionally successful films accounted for a large portion of the revenue, while most films operated on a moderate market.

This reflects an industry in which success was less predictable and large-scale investments were relatively rare.

In contrast, the second plot shows a significantly stronger positive correlation between production budgets and revenues, with a steeper regression line and a more structured distribution of data points.

The dense clustering of data points in the mid-to-high budget range suggests that larger investments became more common and were more consistently associated with higher revenues. While there are still outliers, the confidence interval is smaller relative to the sample size, suggesting a more stable and predictable relationship between budget and revenue.

Overall, this comparison illustrates how the modern film industry has evolved into a capital-intensive model in which higher budgets are increasingly accompanied by higher box office revenues, whereas in the past, the film industry relied more on occasional surprise hits than on consistent profitability on a large scale.

Linear Regression Analysis of Budget and Revenue

To quantify the relationship between production budgets and global box office revenue, we are going to apply a linear regression model. Unlike visual regression lines, this approach mathematically estimates the extent to which the budget influences box office results.

Linear regression attempts to model the relationship between an explanatory variable (budget) and a response variable (revenue) by fitting the best straight line through the data. This allows us to measure both the direction and the strength of the relationship and to assess the financial impact of rising production budgets.

To perform this analysis, we are going to utilize the *LinearRegression* model from scikit-learn.

```
from sklearn.linear_model import LinearRegression
regression = LinearRegression()
```

Defining Features and Target Variables

To build the regression model, we need to specify the explanatory variable (feature) and the response variable (target variable).

In this case, the production budget is used as the predictor, and global gross revenue serves as the outcome we want to explain.

The analysis is based exclusively on recent films, as the relationship between budget and revenue was weaker and less consistent during the earlier period. By focusing on recent films, the model can better capture the more structured, contemporary financial dynamics of the film industry.

```
# Explanatory Variable (Feature)
x = pd.DataFrame(new_films, columns=['USD_Production_Budget'])

# Response Variable (Target)
y = pd.DataFrame(new_films, columns=['USD_Worldwide_Gross'])

# Find the best-fit line
regression.fit(x, y)
```

Interpreting the Intercept and Coefficient

After fitting the model, we determine the y-intercept and the slope (the coefficient) of the regression line.

The x-axis represents expected revenue when the production budget is zero, while the coefficient indicates the expected increase in revenue for every additional dollar spent on production.

```
# theta 0:
print("\nIntercept of the Relation:")
print(regression.intercept_)

# theta 1:
print("\nRegression Coefficient:")
print(regression.coef_)
```

```
Intercept of the Relation:
[-8528035.18538941]
```

```
Regression Coefficient:
[[3.13050834]]
```

The coefficient of approximately 3.13 suggests that, on average, every additional dollar invested in the production budget is associated with approximately \$3.13 in global gross revenue.

This indicates a strong positive relationship between investment and returns, meaning that higher budgets tend to lead to a significant increase in revenue.

On the other hand, the negative intercept has limited real-world interpretation, since a film with a zero budget is unrealistic; however, it reflects the fact that the model fits the lines in such a way that they best match the observed data.

Overall, the results support earlier visual findings that higher investments in modern film production are closely linked to higher box office revenues.

Estimating the Break-Even Budget

Another useful insight from the regression model is the estimation of the theoretical break-even point. The break-even point is reached when revenue equals the production budget, meaning that the film neither generates a profit nor incurs a loss.

Using the regression equation, we can calculate the budget value at which the projected revenue equals the investment.

$$\text{Budget} = 3.124 \times \text{Budget} - 8,528,035$$

$$8,528,035 = (3.131 - 1) \times \text{Budget}$$

$$8,528,035 = 2.131 \times \text{Budget}$$

$$\text{Budget} = 8,528,035 / 2.131$$

$$\text{Budget} \approx \$4,001,893$$

Based on the adjusted model, the break-even point is approximately \$4 million.

This suggests that, in line with the average trend for modern films, productions that exceed this threshold are more likely to generate a profit.

From this point on, every additional dollar invested in production generates approximately \$3.12 in revenue, which equates to an average profit of \$2.12 for every additional dollar spent.

While this is only a statistical estimate and real-world outcomes vary widely, it highlights how even relatively modest budgets can become profitable when aligned with the overall industry trend.

Making Revenue Predictions Using the Regression Model

Now that we have estimated the intercept and regression coefficient (slope) of the linear regression model, we can use the fitted equation to make revenue forecasts for new films. In this way, we can translate the statistical relationship between the budget and revenue into concrete financial projections. By entering a hypothetical production budget, the model estimates the average global gross revenue based on historical trends for modern films.

As an example, we are going to estimate the expected revenue for a film with a production budget of \$350 million.


```
budget = 350000000
revenue_estimate = regression.intercept_[0] + regression.coef_[0,0]*budget
revenue_estimate = round(revenue_estimate, -6)

print(f'\nThe estimated revenue for a 350 Million Dollars film is around
${revenue_estimate:,}')
```

The estimated revenue for a new film with the budget of \$350 million is around \$1,087,000,000.0

This projection suggests that, based on the historical ratio of budget to revenue, a production with a budget of \$350 million is likely to exceed \$1 Billion on average.

Based on this ratio, this also translates to an estimated profit of approximately \$737 million, illustrating how large-scale productions can generate substantial returns when they follow the general industry trend.

However, this figure represents an average estimate rather than a guarantee and there are many factors that can effect actual results such as brand strength, marketing, audience response, and competition at the time of release. In the next section we are going to investigate how well and accurate our estimation tends to be.

Measuring Model Accuracy with R^2 (Coefficient of Determination)

To assess how well the regression model explains the data, we look at the R^2 value. The R^2 value measures the proportion of variation in the response variable (revenue) that can be explained by the predictor (budget).

A value closer to 1 indicates a stronger explanatory relationship, while a value near 0 suggests weak predictive power.

```
regression.score(x, y)
```

```
>> 0.5581333778099973
```

We see that the coefficient of determination measures approximately 0.56 in our Analysis.

An R^2 value of 0.56 means that about 56% of the variation in global revenue can be explained solely by the production budget.

This is a relatively good result, considering that the model uses only one variable and ignores many other important factors, such as marketing, the franchise's popularity, the genre, the release date, and audience reception.

The remaining 44% of the variation is likely attributable to these additional factors, which explains why some mid-budget films exceed expectations while some high-budget films fall short of them.

Overall, this finding confirms that while budget is a key factor in box office revenue, it is not the only factor determining a film's financial success.

Linear Regression Analysis for Pre-1960 Films

To compare financial trends across different eras, we are going to apply the same regression analysis to films released before 1960. This allows us to examine whether production budgets in early cinema had a similar impact on box office revenues as they do in the modern film industry.

If this correlation is weak, it would suggest that the financial success of earlier films depended more on factors such as cultural impact, long theatrical runs, and low competition than on production costs.

This analysis therefore helps validate the decision to separate old and modern films in previous sections.

Fitting the Regression Model

The regression model is fitted using production budget as the explanatory variable and worldwide gross revenue as the response variable.

```
x = pd.DataFrame(old_films, columns=['USD_Production_Budget'])
y = pd.DataFrame(old_films, columns=['USD_Worldwide_Gross'])

# Find the best-fit line
regression.fit(x, y)
```

```
print(f'The Regression (slope) coefficient is: {regression.coef_[0][0]:.3}')  
print(f'The intercept is: {regression.intercept_[0]:.3}')  
print(f'The r-squared is: {regression.score(x, y):.3}')
```

The Regression (slope) coefficient is: 1.54

The intercept is: 2.52

The r-squared is: 0.0258

The extremely low R^2 (≈ 0.03) suggests that the production budget explains only about 3% of the variation in box office revenue for films released before 1960, indicating that there is almost no significant linear relationship.

This suggests that in early cinema, higher expenditure did not reliably lead to higher box office incomes, which is consistent with the scattered pattern observed in the previous regression plot.

Furthermore, the Regression coefficient of 1.54 is significantly lower than that of modern films, underscoring the fact that budget had a lesser impact on revenue generation during that period.

Overall, this finding strongly supports the decision to analyze old and modern films separately, as combining them would obscure the fundamentally different financial structures of the two periods.

Conclusion

This report examined financial trends in the film industry by analyzing production budgets, box office revenues, profits, and profit margins over various time periods. By examining the most successful films, the biggest box office hits, and long-term budget trends, the analysis demonstrated how the relationship between investment and financial success has evolved over time. The comparison between early cinema and modern filmmaking revealed a structural shift toward larger budgets and a stronger correlation between budget and revenue, while also identifying influential outliers that have shaped industry expectations.

From a data science perspective, the analysis combined techniques of data cleaning, transformation, statistical analysis, and visualization to extract meaningful insights from the dataset. The workflow involved processing missing and inconsistent data, creating derived metrics such as profit and profit margin, filtering out non-representative entries, and running conditional queries to identify specific observations. To analyze the data, both static and interactive visualizations were used, including bar charts, 3D scatter plots, and regression plots, to examine the relationships between the variables. Finally, linear regression models and their evaluation using the coefficient of determination (R^2) made it possible to quantify the strength of the relationship between budget size and income, estimate break-even points, and generate revenue forecasts. Overall, the report demonstrates how the combination of data preparation, visualization, and statistical modeling can provide a comprehensive understanding of financial trends in the film industry.

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